FACULDADE DE CIÊNCIAS DA UNIVERSIDADE DO PORTO
FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

Inference-Time Techniques for Open-Weight Language Models in Negotiation Games

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WORKING VERSION

Master in Artificial Intelligence

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June 1, 2026

Resumo

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Palavras-chave: keyword1 • keyword2 • keyword3 • keyword4

Abstract

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The United Nations Sustainable Development Goals

The United Nations Sustainable Development Goalss (SDGs) provide a global framework for achieving a better and more sustainable future for all. It includes 17 goals (`https://sdgs.un.org/goals`) to address the world's most pressing challenges, including poverty, inequality, climate change, environmental degradation, peace, and justice.

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The specific Sustainable Development Goals mentioned have the following names:

SDG 7 Ensure access to affordable, reliable, sustainable and modern energy for all

SDG 8 Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all

SDG	Target	Contribution	Performance Indicators and Metrics
7	7.1	Enhancing the efficiency of SCADA systems improves the reliability and performance of solar energy production, supporting universal access to clean and affordable energy.	Percentage of solar plants with …
	7.2	Improved management of solar plants increases operational efficiency and reliability, contributing to a higher share of renewable energy in the energy mix.	Increase in renewable energy share …
8	8.1	Strengthening renewable energy infrastructure enhances resilience to climate-related hazards and supports sustainable, long-term energy systems.	Improvement in infrastructure …

Acknowledgements

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<The Full Name of the Author>

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List of Acronyms

AI	Artificial Intelligence
FEUP	Faculdade de Engenharia da Universidade do Porto
GPU	Graphics Processing Unit
ML	Machine Learning
NLP	Natural Language Processing
SDG	Sustainable Development Goals

Chapter 1

Introduction

Almost everything in life depends on negotiation. From the moment we are born, we negotiate for attention and care. Developed in part through the unconscious imitation of those around us [12], negotiation requires the ability to infer others' intentions and to act under uncertainty.

Large language models (LLMs) have demonstrated impressive capabilities across a number of tasks, from code generation to creative writing and question answering. As these models are increasingly deployed in agentic scenarios—where they must interact with other agents or humans to accomplish goals—negotiation appears as a particularly interesting benchmark.

Bianchi et al. [2] introduced NegotiationArena, an open-source framework for evaluating the negotiation abilities of LLM agents across scenarios such as the Ultimatum Game, resource trading, and price negotiation. Although highly valuable, the original evaluation was limited to closed-source models (GPT-4, GPT-3.5, Claude 2.1, and Claude 2), some of which are no longer available. This limitation raises concerns regarding reproducibility and accessibility.

Techniques such as Chain-of-Thought (CoT) allow models to think step-by-step, dividing complex problems into manageable sub-problems to reduce errors and improve interpretability. Similarly, Self-Correction mechanisms (or Self-Refine) enable models to critique and improve their own outputs through an iterative process of drafting, feedback, and refinement. While these emergent abilities have demonstrated success in domains like mathematics and coding , their specific impact on negotiation has not yet been fully studied.

Beyond individual reasoning, multi-agent collaboration offers another area for improvement. Cooperative debate, where multiple agents generate, critique, and refine each other's responses, has been shown to improve factual accuracy and reasoning quality. In real-world negotiation, decisions are rarely made by a single individual; corporate acquisitions, policy agreements, and legal settlements typically involve teams that deliberate internally before engaging with the opposing side. This observation motivates the question of whether team-based negotiation — where each side is represented by a group of LLM agents that debate before responding — can lead to stronger negotiation outcomes.

This thesis investigates whether inference-time reasoning techniques and multi-agent collaboration can improve the negotiation capabilities of large language models. The work is organized around three main contributions:

- How do open-source models compare to proprietary models in competitive negotiation scenarios?
- As inference-time reasoning techniques have shown promise individually, to what extent do they affect outcomes in multi-agent negotiation contexts?
- Can team-based negotiation, where each side is represented by a group of LLM agents that deliberate before responding, lead to stronger negotiation outcomes?

The remainder of this document is organized as follows. Chapter 2 provides a literature review covering the evolution of language models, inference-time reasoning techniques, multi-agent systems, and the different interaction scenarios relevant to this work. Chapter 3 presents the detailed work plan for the upcoming months.

Chapter 2

Literature Review

2.1 Language Models

The origins of language modeling date back to Shannon [31], which modeled natural language as a stochastic process. Building on this, Bengio et al. [1] proposed one of the first Neural Language Models (NLMs) estimating n-gram probabilities using neural networks. Thereafter, NLMs based on Long Short-Term Memory (LSTM) became the standard for many natural language tasks, including machine translation, text generation, and classification [24, 34].

The introduction of the Transformer architecture [43] addressed some limitations of LSTMs, allowing parallelization during training and capturing long-range dependencies through the self-attention mechanism. Applying this architecture, Radford and Narasimhan [29] pre-trained a language model on the next-word prediction task and showed that, when followed by task-specific fine-tuning, it achieved state-of-the-art performance on common-sense reasoning, question answering and many other tasks. GPT-2 [30] demonstrated that sufficiently large language models can perform downstream tasks such as summarization, translation, and question answering without task-specific fine-tuning.

The increase in the number of parameters, dataset size, and training compute of language models continued, supported by the power-law relationship detected in Kaplan et al. [18], which showed that an increase in these three components led to an improvement in performance. As models grew, so did the computational costs and risks associated with their deployment, which led the majority of frontier labs to keep the model weights proprietary; nevertheless, some have opted for an open-weight release (e.g., LLaMA [13], Mistral [16], DeepSeek [9]). Table 2.1 compares some of the main families of models.

2.2 Inference-Time Reasoning Techniques

Beyond simple improvements in performance, scaling also brought new capabilities that could not be predicted by extrapolating the performance of smaller models. Wei et al. [45] refers to them as *emergent abilities*. Among them are the ability to learn from examples (Few-Shot prompting,

Table 2.1: Comparison of different models. For Mixture of Experts (MoE) models, parameters are denoted as *Total / Active*.

Model	**Parameters**	**License**	**Notes**
OpenAI			
GPT-1	117M	Proprietary	[29, 30]
GPT-2	117M–1542M (4 variants)	Open-weights	[30]
GPT-3	125M–175B (7 variants)	Proprietary	[4]
GPT-4, GPT-5	Undisclosed	Proprietary	[26, 33]
GPT-OSS	117B/5.1B, 21B/3.6B	Open-weights	MoE [27]
Google			
Gemini 1.0–3	Undisclosed	Proprietary	[8, 35]
Gemma	2B, 7B	Open-weights	[38]
Gemma 2	2B, 9B, 27B	Open-weights	[36]
Gemma 3	1B, 4B, 12B, 27B	Open-weights	Multimodal [37]
Meta			
LLaMA 1	7B, 13B, 33B, 65B	Open-weights	[41]
LLaMA 2	7B, 13B, 70B	Open-weights	[40]
LLaMA 3	8B, 70B, 405B	Open-weights	[13]
LLaMA 4	109B/17B, 400B/17B	Open-weights	MoE [23]
Anthropic			
Claude 3, 4.5	Undisclosed	Proprietary	[39]
DeepSeek			
DeepSeek-R1	671B/37B	Open-weights	MoE [14]

[3]), to think step-by-step (Chain-of-Thought, [44]), and to criticize and improve its own outputs (Self-Refine, [22]).

2.2.1 Chain-of-Thought Reasoning

Depending on how Chain-of-Thought is implemented, it can be classified as Zero-Shot-CoT [20] if we simply attach an instruction that elicits reasoning (e.g., "Let's think step by step") or Few-Shot-CoT [44] if we also accompany it with examples of reasoning traces.

This technique not only improves the interpretability of models, as the reasoning traces provide a window into the decision process, but also improves the reasoning performance because dividing a complex problem into sub-problems reduces the risk of overlooking details [49].

Large reasoning models (LRMs) that internalize Chain-of-Thought (CoT) through reinforcement learning [14, 28] have become state-of-the-art, achieving a significant leap in performance, especially in coding and math capabilities. Shojaee et al. [32] analyzed the advantages of LRMs and identified three different performance regimes. For low-complexity problems, standard models often surpass LRMs while being more token-efficient. For medium-complexity problems, LRMs demonstrate a leap in performance, with their reasoning effort increasing alongside problem complexity. Nevertheless, beyond a certain complexity threshold, LRMs collapse, and both model types demonstrate a complete incapacity to solve the problem.

2.2.2 Self-Correction

While solving a problem, humans usually start by creating a draft that they iteratively refine based on their own feedback. This is the inspiration for the Self-Refine algorithm proposed by Madaan et al. [22] that breaks down the generation process into three steps: (1) the model generates an initial draft, (2) generates feedback, and (3) based on this feedback, refines the answer.

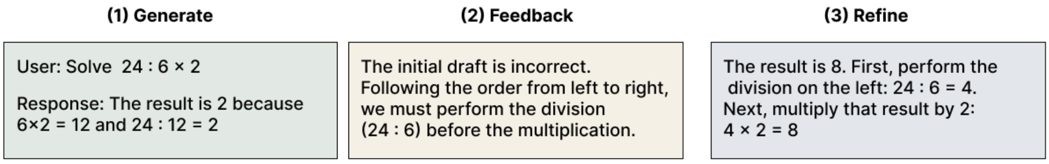

Figure 2.1: Example of the Self-Refine process. The model critiques its initial output to generate an improved final response based on self-generated feedback.

Subsequent work continued this idea of self-correction, applying it to various tasks such as arithmetic reasoning and code generation. Some frameworks provide additional information in the feedback stage, such as the output from a code interpreter [7]. As noted in Kamoi et al. [17], there is conflicting evidence regarding the efficacy of LLM self-correction. Their survey concluded that, although self-correction works well when there is external feedback or the models are fine-tuned for self-correction, prior work has demonstrated success only on tasks exceptionally suited to self-correction.

2.3 Multi-Agent Systems

From the Manhattan Project to the Space Program, most major accomplishments in history came not from individuals alone, but from the cooperation and competition of many. This observation serves as inspiration for the study of multi-agent systems, defined in Wooldridge [46] as systems where agents interact with each other by exchanging messages.

Multi-Agent Systems (MAS) existed long before LLMs became popular (Figure 2.2), dating back to the 1970s with distributed problem solving. In the last decades, deep reinforcement learning and large language models have significantly expanded MAS capabilities [25].

Li et al. [21] proposed a framework for LLM-based agents, identifying the following five components:

- Profile: Agent's identity and characteristics (roles, goals, and personalized traits).
- Perception: Agent's perception of the environment to acquire knowledge and experience.
- Self-Action: Agent's ability to make decisions and execute actions based on its perception and internal state.

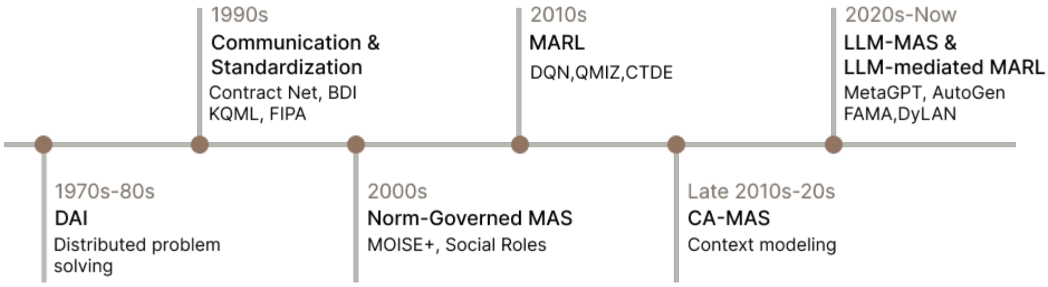

Figure 2.2: Evolution of MAS paradigms. Adapted from Nguyen-Thi-Mai et al. [25].

- Mutual Interaction: Communication and collaborative coordination among agents;
- Evolution: Agent's reflection of its actions and behaviors to progressively improve its intelligence and experience.

2.4 Interaction Scenarios

According to Li et al. [21], interaction scenarios in LLM-Based Multi-Agent systems can be classified as cooperative, adversarial, or mixed.

2.4.1 Cooperative

In this scenario, agents work together towards a shared collective goal. It usually follows these steps: after setting a common goal, agents decompose the task and use communication and negotiation to reach a consensus [42].

Du et al. [10] introduces collaborative debate as a way to improve the factual validity and the mathematical and strategic reasoning across a number of tasks. First, a set of LLMs generates possible answers to a question. Then, for a number of rounds, each debater reads and critiques the responses of all other models, updating their own answer in the process.

Inspired by human group dynamics, Chen et al. [6] proposes AgentVerse, a framework for problem-solving. Its process is divided into four steps. First, a recruiter agent decomposes the task by writing a set of expert descriptions for each agent. Then, the selected agents collaborate to decide on the actions to take. This communication can have a horizontal structure, where all agents share and integrate their decisions to decide the action to take (often used for consulting/tool use), or a vertical structure, where a solver agent proposes an initial solution and a set of reviewers critique it to drive iterative refinement (ideal for math/software development tasks). Finally, the action is executed, and feedback is provided by assessing the difference between the current state and the desired goal.

2.4.2 Adversarial

Competition occurs when agents have conflicting objectives and seek to maximize their own interests, often with limited resources [21, 42].

2.4.2.1 Debate

Inspired by Irving, Christiano, and Amodei [15]'s idea of debate as an approach to AI alignment, Khan et al. [19] implements an information-asymmetric debate where two expert models (debaters) argue for different sides over N rounds, attempting to persuade a weaker model (judge) that must identify the correct side. This work has the objective of answering the question "can weaker models assess the correctness of stronger models?", motivated by a future where humans need to supervise more powerful models.

Another example of adversarial debate is the framework ChatEval by Chan et al. [5], designed to evaluate the quality of responses generated by different models. At the end of the debate, instead of asking the debater agents to reach a consensus, they derive the final results through majority vote.

2.4.2.2 Game-Theoretic Scenarios

Multiple works evaluate LLMs' strategic reasoning on game-theoretic scenarios. Feng et al. [11] distinguishes two types of games. Choice-focused if there is little to no communication between the participants (e.g., Poker [47]) or communication-focused when communication is the main component of the game (e.g., Negotiation [2]).

Zhan et al. [48] define negotiation as the process by which two or more parties attempt to resolve their opposing interests. Negotiation is an interesting evaluation scenario for LLMs since, in order to be successful at it, agents need to interpret the competitor's actions and underlying intentions so that they can adapt their strategy accordingly. Bianchi et al. [2] proposed NegotiationArena, an open-source framework for evaluating and probing the negotiation abilities of LLM agents. In it, they evaluated three scenarios:

- Ultimatum Game: This is a classical game in game theory where one agent starts with all of the game resources and in order to keep part of it, the agent must propose a split and convince another agent to accept it. If the agent rejects it, the game is over and they both get nothing. Bianchi et al. [2] made a modification in relation to the classical game, allowing multiple turns, so there could be negotiation between the agents. The agent who secures more resources is considered the winner.

- Trading Game: In this game, each agent has a set of resources and a goal (e.g., maximize its total resources). For a number of rounds, the agents make proposals to each other until one accepts a proposal. Similar to the Ultimatum Game, the agent who obtains more resources is considered the winner.

- Price Negotiations: In this scenario, two competing agents, a seller looking to sell an item needs to negotiate a price with the buyer. This is an incomplete information game since each agent has private information (the seller's cost and the buyer's willingness to pay), which is not directly observable by the other party. An agent is considered a winner if the final selling point is closer to their opponent's reservation price (the cost for the seller or the willingness to pay for the buyer) than their own.

The framework used two types of metrics: win rate and average payoff. By analyzing them, we can get meaningful comparisons of the models' ability to negotiate with others.

2.4.3 Mixed

Some scenarios combine features of both cooperative and adversarial interactions. Li et al. [21] further divided these scenarios into two types:

- Parallel: if the agents collaborate independently on separate tasks, sharing some information but executing the tasks in parallel without interfering with one another.
- Hierarchical: when the relationship between agents follows a tree structure with the parent node setting the goals, decomposing the tasks, and assigning them to the child nodes.

Chapter 3

Chapter WITH Examples

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3.1 Introduction

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[1] https://en.wikipedia.org/wiki/Equation

[2] https://www.overleaf.com/learn/latex/Mathematical_expressions

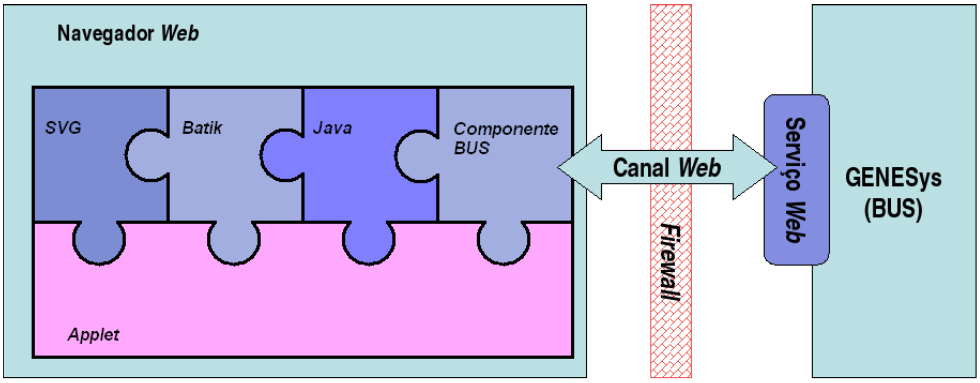

Figure 3.1: A figure, caption below

3.2 Section Example

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- **Praesent** — Sit amet sem maecenas eleifend facilisis leo;
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3.2.1 Subsection Example — Figures

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3.2.2 Subsection Example — Tables

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[3] `https://tex.stackexchange.com/questions/381426/advantage-of-floating-figures`

Table 3.1: A table, caption on top

Iteration k de $f(x_n)$				
$\boldsymbol{k}$	x_1^k	x_2^k	x_3^k	comments
0	-0.3	0.6	0.7	-
1	0.47102965	0.04883157	-0.53345964	$\delta < \varepsilon$
2	0.49988691	0.00228830	-0.52246185	$\delta < \varepsilon$
3	0.49999976	0.00005380	-0.523656	N
4	0.5	0.00000307	-0.52359743	
$\vdots$	$\vdots$	$\ddots$	$\vdots$	
7	0.5	0.0	**-0.52359878**	$\delta < 10^{-8}$

placerat fermentum, augue nisl ultricies eros, id fringilla enim sapien eu felis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Sed dolor mi, porttitor quis, condimentum sed, luctus in.

3.2.3 Subsection Example — SI

Always use the `siunitx` package to write values, units, etc. according to the International System of Units convention rules and style conventions (see `https://physics.nist.gov/cuu/Units/checklist.html`).

For example: 50 kg, 15 °C, 20.01 m s^{-1}, 120 kB s^{-1}, 34 %, 289 300.00 €, 10 245.450 234.

Table 3.2 serves to exemplify the alignment in the columns with numbers, and has nothing to do with any numerical data associated with FEUP's resources and investments in 2025.

Loren ipsum dolor sit amet, consectetuer adipiscing elit. Praesent sit amet sem. Maecenas eleifend facilisis leo. Vestibulum et mi. Aliquam posuere, ante non tristique consectetuer, dui elit scelerisque augue, eu vehicula nibh nisi ac est. Suspendisse elementum sodales felis. Nullam laoreet fermentum urna.

Duis eget diam. In est justo, tristique in, lacinia vel, feugiat eget, quam. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Fusce feugiat, elit ac placerat fermentum, augue nisl ultricies eros, id fringilla enim sapien eu felis. Vestibulum ante

Table 3.2: Physical Resources of FEUP [**kn:feup**].

Some Values
2.3456
34.2345
-6.7835
90.473
5642.5
1.2×10^3
10^4

ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Sed dolor mi, porttitor quis, condimentum sed, luctus in[4].

3.2.4 Subsection Example — Acronyms

This section shows how to use the acronyms Artificial Intelligence (AI) and Machine Learning (ML). Later we can simply refer to AI or ML without expansion.

In modern technology, Natural Language Processing (NLP) plays a crucial role in virtual assistants and chatbots. We also rely on Graphics Processing Unit (GPU) hardware for deep learning models.

3.2.4.1 Special use in titles as FEUP

Acronyms can also be made plural: GPUs are widely used in high-performance computing. Possessive forms are supported too: AI's impact on society is profound at Faculdade de Engenharia da Universidade do Porto (FEUP).

3.3 Summary

Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Fusce feugiat, elit ac placerat fermentum, augue nisl ultricies eros, id fringilla enim sapien eu felis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Sed dolor mi, porttitor quis, condimentum sed, luctus in.

[4]The `siunitx` package Manual is available at `https://ctan.org/pkg/siunitx`.

Chapter 4

Another chapter

Integer nec quam. Sed fermentum. Nunc vitae leo. Etiam sit amet quam. Nunc vestibulum massa in mauris. Duis eget nulla.

4.1 Section Example

Fusce ultricies arcu eu nibh volutpat feugiat. Maecenas urna pede, commodo quis, porta eu, bibendum elementum, pede.

4.2 Section Example

Sed eros massa, molestie eget, mattis non, rutrum ac, magna. Duis dui. Maecenas eget tortor ut dolor semper mattis. Maecenas auctor, tellus et ultricies tempor, elit est placerat lacus, in posuere mauris lorem et arcu.

4.2.1 Subsection Example

Nulla nec eros et pede vehicula aliquam. Aenean sodales pede vel ante. Fusce sollicitudin sodales lacus. Maecenas justo mauris, adipiscing vitae, ornare quis, convallis nec, eros.

4.2.2 Subsection Example

Pellentesque pulvinar fringilla dolor. In sit amet pede. Proin orci justo, semper vel, vulputate quis, convallis ac, nulla. Nulla at justo. Mauris feugiat dolor. Etiam posuere fermentum eros. Morbi nisl ipsum, tempus id, ornare quis, mattis id, dolor. Aenean molestie metus suscipit dolor. Aliquam id lectus sed nisl lobortis rhoncus. Curabitur vitae diam sed sem aliquet tempus. Sed scelerisque nisi nec sem.

4.3 Section Example

Sed eros massa, molestie eget, mattis non, rutrum ac, magna. Duis dui. Maecenas eget tortor ut dolor semper mattis. Maecenas auctor, tellus et ultricies tempor, elit est placerat lacus, in posuere mauris lorem et arcu.

4.3.1 Subsection Example

Nulla nec eros et pede vehicula aliquam. Aenean sodales pede vel ante. Fusce sollicitudin sodales lacus. Maecenas justo mauris, adipiscing vitae, ornare quis, convallis nec, eros.

4.3.2 Subsection Example

khakipoor_linear_2023, **liu_energy_2023** aliquam id lectus sed nisl lobortis rhoncus. Curabitur vitae diam sed sem aliquet tempus. Sed scelerisque nisi nec sem.

4.4 Section Example

Maecenas urna pede, commodo quis, porta eu, bibendum elementum, pede. Sed eros massa, molestie eget, mattis non, rutrum ac, magna. Duis dui. Maecenas eget tortor ut dolor semper mattis. Maecenas auctor, tellus et ultricies tempor, elit est placerat lacus, in posuere mauris lorem et arcu [**monopoli_exploiting_2023**, **zhang_carma_2023**, **chang_adas_2023**, **guo_rapidstream_2023**].

4.4.1 Subsection Example

Nulla nec eros et pede vehicula aliquam. Aenean sodales pede vel ante. Fusce sollicitudin sodales lacus. Maecenas justo mauris, adipiscing vitae, ornare quis, convallis nec, eros.

4.4.2 Subsection Example

khakipoor_linear_2023, **liu_energy_2023** aliquam id lectus sed nisl lobortis rhoncus. Curabitur vitae diam sed sem aliquet tempus. Sed scelerisque nisi nec sem scelerisque.

Nulla nec eros et pede vehicula aliquam. Aenean sodales pede vel ante. Fusce sollicitudin sodales lacus. Maecenas justo mauris, adipiscing vitae, ornare quis, convallis nec, eros. Etiam laoreet venenatis ipsum. In tellus odio, eleifend ac, ultrices vel, lobortis sed, nibh. Fusce nunc augue, dictum non, pulvinar sed, consectetuer eu, ipsum. Vivamus nec pede.

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Appendix A

Lorem Ipsum

After the conclusions and bibliographical references, the text used to complete the dissertation is presented in this numbered annex.

A.1 What is *Lorem Ipsum*?

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum [**kn:Lip08**].

A.2 Where does *Lorem* come from?

Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature from 45 BC, making it over 2000 years old. Richard McClintock, a Latin professor at Hampden-Sydney College in Virginia, looked up one of the more obscure Latin words, consectetur, from a Lorem Ipsum passage, and going through the cites of the word in classical literature, discovered the undoubtable source. Lorem Ipsum comes from sections 1.10.32 and 1.10.33 of "de Finibus Bonorum et Malorum" (The Extremes of Good and Evil) by Cicero, written in 45 BC. This book is a treatise on the theory of ethics, very popular during the Renaissance. The first line of Lorem Ipsum, "Lorem ipsum dolor sit amet...", comes from a line in section 1.10.32.

The standard chunk of Lorem Ipsum used since the 1500s is reproduced below for those interested. Sections 1.10.32 and 1.10.33 from "de Finibus Bonorum et Malorum" by Cicero are also reproduced in their exact original form, accompanied by English versions from the 1914 translation by H. Rackham.

A.3 Why is *Lorem* used?

It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters, as opposed to using "Content here, content here", making it look like readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for "lorem ipsum" will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humour and the like).

A.4 Where can you find examples?

There are many variations of passages of Lorem Ipsum available, but the majority have suffered alteration in some form, by injected humour, or randomised words which don't look even slightly believable. If you are going to use a passage of Lorem Ipsum, you need to be sure there isn't anything embarrassing hidden in the middle of text. All the Lorem Ipsum generators on the Internet tend to repeat predefined chunks as necessary, making this the first true generator on the Internet. It uses a dictionary of over 200 Latin words, combined with a handful of model sentence structures, to generate Lorem Ipsum which looks reasonable. The generated Lorem Ipsum is therefore always free from repetition, injected humour, or non-characteristic words etc.